

Elixir style: source comparison

Generated with Elixir 1.19.5. A guide column shows "Yes" where it proposes the rule; the mix format column shows "Enforced" where the formatter applies it automatically. 114 master rules.
Sources a mix format = mix format; Lexmag = Lexmag's Elixir Style Guide; Credo = Credo's Elixir Style Guide; Adams = The Elixir Style Guide.

Enforced by mix format

Rule	mix format	Lexmag	Credo	Adams
Avoid aligning expression groups	Enforced	Yes		
Binary operators at end of line	Enforced	Yes		
Blank line after a multi-line assignment	Enforced			Yes
Blank lines around multi-line case clauses	Enforced			Yes
Capture operator spacing	Enforced			
Collapse consecutive blank lines	Enforced			
Comments on their own line	Enforced			Yes
Fit collections on a single line when they fit	Enforced			
for clause indentation	Enforced	Yes		
Format with/else over multiple lines	Enforced			Yes
Group digits in large numbers	Enforced	Yes	Yes	
Guard clause indentation	Enforced	Yes		
Indent multi-line binary operands	Enforced	Yes		
Lowercase exponent letter in floats	Enforced			
Maximum line length	Enforced		Yes	Yes
Multi-line expression on its own line after =	Enforced	Yes	Yes	
No blank line after defmodule	Enforced			Yes
No parentheses around anonymous function arguments	Enforced	Yes		
No redundant parentheses	Enforced			
No semicolons between statements	Enforced	Yes	Yes	
No space after the ! operator	Enforced	Yes	Yes	Yes
No space between a function name and (	Enforced			Yes
No spaces around bitstring segment options	Enforced	Yes		
No spaces around the range operator	Enforced	Yes		Yes
No spaces inside brackets	Enforced		Yes	Yes
No trailing comma in collections	Enforced	Yes		
No trailing whitespace	Enforced	Yes	Yes	Yes
Omit brackets from a defstruct keyword list	Enforced			Yes
Omit brackets from keyword lists	Enforced			Yes
One element per line in multi-line collections	Enforced			Yes
One part per line in long union types	Enforced			Yes
Parentheses around definition arguments	Enforced	Yes	Yes	
Parentheses on function calls with arguments	Enforced		Yes	Yes

Formatting

Rule	mix format	Lexmag	Credo	Adams
Long do: clause on its own line				Yes
Opening bracket stays on the assignment line				Yes
Use vertical space between definitions			Yes	Yes

Naming

Rule	mix format	Lexmag	Credo	Adams
Avoid one-letter variable names		Yes		
CamelCase for modules		Yes	Yes	Yes
Don't shadow Kernel functions			Yes	
is_ prefix for guard-safe predicates				Yes
Private function names differ from public				Yes
Question mark for predicate functions		Yes	Yes	Yes
snake_case for atoms, variables and functions		Yes	Yes	Yes
snake_case for files and directories		Yes		Yes

Modules

Rule	mix format	Lexmag	Credo	Adams
Alias a long module self-reference				Yes
Alias the modules you use			Yes	
Avoid repetitive module names				Yes
Consistent def line style				Yes
Consistent module attribute layout		Yes		Yes
Don't shadow stdlib modules			Yes	
Group clauses of the same function			Yes	Yes
Module name mirrors its file path				Yes
No explicit nil struct field defaults		Yes		Yes
One module per file				Yes
Order alias/import/require alphabetically				Yes
Use the __MODULE__ pseudo-variable		Yes	Yes	Yes

Typespecs

Rule	mix format	Lexmag	Credo	Adams
@spec directly before the function				Yes
Keep @typedoc with its @type				Yes
Name the main type t				Yes
Parentheses on zero-arity types		Yes		

Expressions

Rule	mix format	Lexmag	Credo	Adams
Avoid needless metaprogramming				Yes
Avoid needless pipelines		Yes		Yes
Don't deeply nest conditionals			Yes	
No always-true conditionals			Yes	
No anonymous functions in pipelines		Yes		
No else clause with unless		Yes	Yes	Yes
No IEx.pry in production code			Yes	
No IO.inspect in production code			Yes	
No parentheses around if/unless conditions			Yes	
No unless with a negated condition			Yes	
Omit else that returns nil		Yes		
Omit parentheses on zero-argument definitions				Yes
Shorthand map syntax for atom keys				Yes
Start pipe chains with a raw value			Yes	Yes
Use <> to match binaries		Yes		Yes
Use and/or/not for boolean checks		Yes		
Use do: for single-line if/unless				Yes
Use keyword-list shorthand				Yes
Use pipelines for chained calls		Yes		Yes
Use true as the last cond clause		Yes		Yes
Verbose map syntax for non-atom keys				Yes

Documentation

Rule	mix format	Lexmag	Credo	Adams
@moduledoc false when undocumented			Yes	Yes
Always include @moduledoc				Yes
Blank line after @moduledoc			Yes	Yes
Use heredocs for documentation				Yes

Comments

Rule	mix format	Lexmag	Credo	Adams
Avoid superfluous comments		Yes		
Capitalise and punctuate comments				Yes
Comments only for important details		Yes		
Limit comment lines to 100 characters				Yes
Prefer @doc/@moduledoc over comments			Yes	
Uppercase annotation keywords (TODO/FIXME/âf)			Yes	Yes

Testing

Rule	mix format	Lexmag	Credo	Adams
Expected value on the right in assertions		Yes		Yes

Exceptions

Rule	mix format	Lexmag	Credo	Adams
Error suffix for exception modules		Yes	Yes	Yes
Lowercase exception messages		Yes		Yes
No trailing punctuation in exception messages		Yes		Yes

Regex

Rule	mix format	Lexmag	Credo	Adams
Prefer pattern matching over regular expressions		Yes		
Use \A and \z for string boundaries		Yes	Yes	
Use non-capturing regex groups		Yes		
Use the ~r sigil for regexes			Yes	

Examples

Enforced by mix format

Avoid aligning expression groups

```
# bad
a   = 1
bbb = 2

# good
a = 1
bbb = 2
```

Binary operators at end of line

```
# bad
value = aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa and bbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb and
cccccccccccccccccccccccccccccccccccccccccc

# good
value =
aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa and bbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb and
cccccccccccccccccccccccccccccccccccccccccc
```

Blank line after a multi-line assignment

```
# bad
x =
foo()
|> bar()
y = 1

# good
x =
foo()
|> bar()
y = 1
```

Blank lines around multi-line case clauses

```
# bad
case x do
1 -> a
2 ->
b
end

# good
case x do
1 ->
a
2 ->
b
end
```

Capture operator spacing

```
# bad
&(&1)
# good
& &1
```

Collapse consecutive blank lines

```
# bad
a = 1
b = 2
# good
a = 1
b = 2
```

Comments on their own line

```
# bad
x = 1 # set x
# good
# set x
x = 1
```

Fit collections on a single line when they fit

```
# bad
[1,
2]
# good
[1, 2]
```

for clause indentation

```
# bad
for a <- as,
b <- bs do
{a, b}
end
# good
for a <- as,
b <- bs do
{a, b}
end
```

Format with/else over multiple lines

```
# bad
with {:ok, x} <- f() do
x
else _ -> :error end
# good
with {:ok, x} <- f() do
x
else
_ -> :error
end
```

Group digits in large numbers

```
# bad
1000000
# good
1_000_000
```

Guard clause indentation

```
# bad
def f(x)
when x > 0 do
x
end
# good
def f(x)
when x > 0 do
x
end
```

Indent multi-line binary operands

```
# bad
x =
foo() +
bar()
# good
x =
foo() +
bar()
```

Lowercase exponent letter in floats

```
# bad
1.0E3
# good
1.0e3
```

Maximum line length

```
# bad
result = some_function(argument_one, argument_two, argument_three, argument_four, argument_five, argument_six)
# good
result =
some_function(
argument_one,
argument_two,
argument_three,
argument_four,
argument_five,
argument_six
)
```

Multi-line expression on its own line after =

```
# bad
x = foo()
|> bar()
# good
x =
foo()
|> bar()
```

No blank line after defmodule

```
# bad
defmodule A do
def x, do: 1
end
# good
defmodule A do
def x, do: 1
end
```


No parentheses around anonymous function arguments

```
# bad
fn(x) -> x end
# good
fn x -> x end
```

No redundant parentheses

```
# bad
(1 + 2)
# good
1 + 2
```

No semicolons between statements

```
# bad
a = 1; b = 2
# good
a = 1
b = 2
```

No space after the ! operator

```
# bad
! ready
# good
!ready
```

No space between a function name and (

```
# bad
foo (x)
# good
foo(x)
```

No spaces around bitstring segment options

```
# bad
<<1 ::8>>
# good
<<1::8>>
```

No spaces around the range operator

```
# bad
1 .. 2
# good
1..2
```

No spaces inside brackets

```
# bad
[ 1, 2 ]
# good
[1, 2]
```

No trailing comma in collections

```
# bad
[1, 2, ]
# good
[1, 2]
```

No trailing whitespace

```
# bad
x = 1
# good
x = 1
```

Omit brackets from a defstruct keyword list

```
# bad
defstruct [name: nil, age: 0]
# good
defstruct name: nil, age: 0
```

Omit brackets from keyword lists

```
# bad
foo([a: 1])
# good
foo(a: 1)
```

One element per line in multi-line collections

```
# bad
[
1, 2,
3
]
# good
[
1,
2,
3
]
```

One part per line in long union types

```
# bad
@type t :: :aaaaa | :bbbbbb | :ccccc | :dddddd | :eeeeee | :ffffff | :ggggg | :hhhhh | :iiii | :jjjj |
# good
@type t ::
:aaaaa
| :bbbbbb
| :ccccc
| :dddddd
| :eeeeee
| :ffffff
| :ggggg
| :hhhhh
| :iiii
| :jjjj
| :kkkkk
```

Parentheses around definition arguments

```
# bad
def add a, b do
a + b
end
# good
def add(a, b) do
a + b
end
```

Parentheses on function calls with arguments

```
# bad
rem 10, 3
# good
rem(10, 3)
```

Parentheses on piped one-arity calls

```
# bad
x |> Enum.reverse
# good
x |> Enum.reverse()
```

Parentheses on zero-arity function calls

```
# bad
DateTime.utc_now
# good
DateTime.utc_now()
```

Quote atoms only when required

```
# bad
:"valid"
# good
:valid
```

Single-level pipeline indentation

```
# bad
x
|> f()
|> g()
# good
x
|> f()
|> g()
```

Space after commas

```
# bad
[1,2]
# good
[1, 2]
```

Space after the comment marker

```
# bad
#comment
# good
# comment
```

Space before -> in 0-arity anonymous functions

```
# bad
fn-> :ok end
# good
fn -> :ok end
```

Spaces around binary operators

```
# bad
1+1
# good
1 + 1
```

Spaces around the -> operator

```
# bad
fn x->x end
# good
fn x -> x end
```

Spaces around the default-argument operator

```
# bad
def f(x \\0), do: x
# good
def f(x \\ 0), do: x
```

Trailing newline at end of file

```
# bad
(file ends without a newline)
# good
(file ends with one \n)
```

Two-space indentation

```
# bad
if x do
:ok
end
# good
if x do
:ok
end
```

Unix line endings

```
# bad
a = 1\r\nb = 2
# good
a = 1\nb = 2
```

Uppercase hexadecimal literals

```
# bad
0xabcd
# good
0xABCD
```

with clause indentation

```
# bad
with {:ok, a} <- f(),
{:ok, b} <- g() do
a + b
end
# good
with {:ok, a} <- f(),
{:ok, b} <- g() do
a + b
end
```

Formatting

Long do: clause on its own line

```
# bad
def handle(arg), do: a_rather_long_expression_that_overflows(arg)
# good
def handle(arg),
do: a_rather_long_expression_that_overflows(arg)
```

Opening bracket stays on the assignment line

```
# bad
x =
[
1,
2
]
# good
x = [
1,
2
]
```

Use vertical space between definitions

```
# bad
def a, do: 1
def b, do: 2
# good
def a, do: 1
def b, do: 2
```

Naming

Avoid one-letter variable names

```
# bad
u = get_user()
# good
user = get_user()
```

CamelCase for modules

```
# bad
defmodule My_app do
end
# good
defmodule MyApp do
end
```

Don't shadow Kernel functions

```
# bad
def length(list), do: count(list)
# good
def list_length(list), do: count(list)
is_prefix for guard-safe predicates
```

```
# bad
defguard valid?(x) when x > 0
# good
defguard is_valid(x) when x > 0
```

Private function names differ from public

```
# bad
def f(x), do: x
defp f(x, y), do: x + y
# good
def f(x), do: x
defp do_f(x, y), do: x + y
```

Question mark for predicate functions

```
# bad
def is_valid(x), do: true
# good
def valid?(x), do: true
```

snake_case for atoms, variables and functions

```
# bad
fooBar = 1
# good
foo_bar = 1
```

snake_case for files and directories

```
# bad
lib/MyApp/UserProfile.ex
# good
lib/my_app/user_profile.ex
```

Modules

Alias a long module self-reference

```
# bad
%MyApp.Deeply.Nested.Thing{}
# good
alias __MODULE__, as: Thing
%Thing{}
```

Alias the modules you use

```
# bad
MyApp.Accounts.User.changeset(u)
# good
alias MyApp.Accounts.User
User.changeset(u)
```

Avoid repetitive module names

```
# bad
MyApp.User.UserHelper
# good
MyApp.User.Helper
```

Consistent def line style

```
# bad
def f(0), do: 0
def f(n) do
  n + 1
end
# good
def f(0) do
  0
end
def f(n) do
  n + 1
end
```

Consistent module attribute layout

```
# bad
alias App.X
use GenServer
@moduledoc "..."/>

```

Module name mirrors its file path

```
# bad
lib/parser.ex defining MyApp.Lexer.Parser
# good
lib/my_app/lexer/parser.ex defining MyApp.Lexer.Parser
```

No explicit nil struct field defaults

```
# bad
defstruct name: nil, age: nil
# good
defstruct [:name, :age]
```

One module per file

```
# bad
defmodule A do
end
defmodule B do
end
# good
```

A and B in separate files

Order alias/import/require alphabetically

```
# bad
alias App.Zebra
alias App.Apple
# good
alias App.Apple
alias App.Zebra
```

Use the `__MODULE__` pseudo-variable

```
# bad
%MyApp.User{name: n}
# good
%__MODULE__{name: n}
```

Typespecs

@spec directly before the function

```
# bad
@spec f() :: :ok
def f, do: :ok
# good
@spec f() :: :ok
def f, do: :ok
```

Keep @typedoc with its @type

```
# bad
@type t :: map()
# â flaterâ f
@typedoc "a thing"
# good
@typedoc "a thing"
@type t :: map()
```

Name the main type t

```
# bad
@type user :: map()
# good
@type t :: map()
```

Parentheses on zero-arity types

```
# bad
@type t :: foo
# good
@type t :: foo()
```

Expressions

Avoid needless metaprogramming

```
# bad
defmacro add(a, b), do: quote(do: unquote(a) + unquote(b))
# good
def add(a, b), do: a + b
```

Avoid needless pipelines

```
# bad
x |> f()
# good
f(x)
```

Don't deeply nest conditionals

```
# bad
if a do
  if b, do: c
end
# good
if a and b, do: c
```

No always-true conditionals

```
# bad
if true do
  go()
end
# good
go()
```

No anonymous functions in pipelines

```
# bad
x |> (fn v -> v + 1 end).()
# good
x |> increment()
```

No else clause with unless

```
# bad
unless x do
  a
else
  b
end
# good
if x do
  b
else
  a
end
```

No IEx.pry in production code

```
# bad
def f(x) do
  IEx.pry()
x
end
# good
def f(x) do
  x
end
```


No IO.inspect in production code

```
# bad
value |> IO.inspect() |> use()
# good
use(value)
```

No parentheses around if/unless conditions

```
# bad
if(ready) do
  go()
end
# good
if ready do
  go()
end
```

No unless with a negated condition

```
# bad
unless not ready do
  go()
end
# good
if ready do
  go()
end
```

Omit else that returns nil

```
# bad
if x, do: a, else: nil
# good
if x, do: a
```

Omit parentheses on zero-argument definitions

```
# bad
def go(), do: :ok
# good
def go, do: :ok
```

Shorthand map syntax for atom keys

```
# bad
%{:a => 1, :b => 2}
# good
%{a: 1, b: 2}
```

Start pipe chains with a raw value

```
# bad
Repo.all(query) |> Enum.map(&f/1)
# good
query |> Repo.all() |> Enum.map(&f/1)
```

Use <> to match binaries

```
# bad
<<"foo", rest::binary>> = str
# good
"foo" <> rest = str
```

Use and/or/not for boolean checks

```
# bad
ready && ok
# good
ready and ok
```

Use do: for single-line if/unless

```
# bad
if x do
  a
end
# good
if x, do: a
```

Use keyword-list shorthand

```
# bad
[{:a, 1}, {:b, 2}]
# good
[a: 1, b: 2]
```

Use pipelines for chained calls

```
# bad
f(g(h(x)))
# good
x |> h() |> g() |> f()
```

Use true as the last cond clause

```
# bad
cond do
  x -> 1
  :else -> 2
end
# good
cond do
  x -> 1
  true -> 2
end
```

Verbose map syntax for non-atom keys

```
# bad
%{"b" => 2, a: 1}
# good
%{"b" => 2, :a => 1}
```

Documentation

@moduledoc false when undocumented

```
# bad
defmodule Internal do
  end
# good
defmodule Internal do
  @moduledoc false
end
```

Always include @moduledoc

```
# bad
defmodule X do
  def f, do: 1
end
# good
defmodule X do
  @moduledoc "â f"
  def f, do: 1
end
```

Blank line after @moduledoc

```
# bad
@moduledoc "X"
def f, do: 1
# good
@moduledoc "X"
def f, do: 1
Use heredocs for documentation
# bad
@doc "line one\nline two"
# good
@doc """
line one
line two
"""
```

Comments

Avoid superfluous comments

```
# bad
# increment counter
counter = counter + 1
# good
counter = counter + 1
```

Capitalise and punctuate comments

```
# bad
# round the total
# good
# Round the total.
```

Comments only for important details

```
# bad
# add one to x
x = x + 1
# good
# round half-up for legacy invoices
x = round_up(x)
```

Limit comment lines to 100 characters

```
# bad
# one very long comment line that runs well past the hundred-character limit and keeps going
# good
# A shorter comment,
# wrapped across lines.
```

Prefer @doc/@moduledoc over comments

```
# bad
# Adds two numbers
def add(a, b), do: a + b
# good
@doc "Adds two numbers"
def add(a, b), do: a + b
```

Uppercase annotation keywords (TODO/FIXME/âf)

```
# bad
# fix this later
# good
# FIXME: handle the timeout case
```

Testing

Expected value on the right in assertions

```
# bad
assert 200 == status
# good
assert status == 200
```

Exceptions

Error suffix for exception modules

```
# bad
defmodule App.BadInput do
end
# good
defmodule App.BadInputError do
end
```

Lowercase exception messages

```
# bad
raise "Invalid input"
# good
raise "invalid input"
```

No trailing punctuation in exception messages

```
# bad
raise "invalid input."
# good
raise "invalid input"
```

Regex

Prefer pattern matching over regular expressions

```
# bad
Regex.match?(~r/^foo/, s)
# good
match?("foo" <> _, s)
```

Use \A and \z for string boundaries

```
# bad
~r/^foo$/
# good
~r/\Afoo\z/
```

Use non-capturing regex groups

```
# bad
~r/(foo|bar)/
# good
~r/(? :foo|bar)/
```

Use the ~r sigil for regexes

```
# bad
Regex.compile!("foo")
# good
~r/foo/
```